

Grid-Forming Control Mode Implementation in PSCAD V5.0.2

PCS 3440 — 3.44 MW / 690 V Power Conversion System

System Rating: 3.44 MVA / 690 V (L-L) / 50/60 Hz

Control Modes: Grid-Forming (VSM + Droop), Grid-Following

Software: PSCAD V5.0.2 (EMTDC Simulation Engine)

Standard: IEEE 1547-2018 / IEC 62786 / IEEE 2800

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PART 1 — SYSTEM SPECIFICATIONS AND BASE VALUES

Part 1: System Specifications and Base Values

1.1 Per-Unit System Base Value Calculations

Per-unit (pu) normalization is the cornerstone of power system analysis and control design. It eliminates the need to track absolute physical quantities across different voltage levels and simplifies the comparison of system parameters. For the PCS 3440, the base quantities are selected to match the rated nameplate values of the converter, following IEEE Std 1459-2010 and IEEE Std 1547-2018 conventions.

1.1.1 Selection of Base Quantities

The three-phase power system requires two independent base quantities — base apparent power S_{base} and base line-to-line voltage $V_{base,LL}$ — from which all other base quantities are derived. The PCS 3440 nameplate ratings directly define these:

Base Quantity Selection (IEEE Convention)

$$S_{base} = 3.44 \text{ MVA} = 3,440,000 \text{ VA}$$

$$V_{base,LL} = 690 \text{ V (line-to-line, RMS)}$$

1.1.2 Derived Base Values

All remaining base quantities are uniquely determined from the two primary selections above. Each derivation is shown in full below.

Base Phase Voltage (Line-to-Neutral):

$$V_{base,LN} = \frac{V_{base,LL}}{\sqrt{3}} = \frac{690}{\sqrt{3}} = \frac{690}{1.7321} = 398.37 \text{ V}$$

Base Three-Phase Current:

$$I_{base} = \frac{S_{base}}{\sqrt{3} \cdot V_{base,LL}} = \frac{3,440,000}{\sqrt{3} \times 690} = \frac{3,440,000}{1,195.3} = 2879.0 \text{ A}$$

Base Impedance (referenced to 690 V side):

$$Z_{base} = \frac{V_{base,LL}^2}{S_{base}} = \frac{690^2}{3,440,000} = \frac{476,100}{3,440,000} = 0.13840 \, \Omega$$

Base Angular Frequency (50 Hz system):

$$\omega_{base} = 2\pi f_{base} = 2\pi \times 50 = 314.159 \, \text{rad/s}$$

Base Inductance:

$$L_{base} = \frac{Z_{base}}{\omega_{base}} = \frac{0.13840}{314.159} = 4.404 \times 10^{-4} \, \text{H} = 0.4404 \, \text{mH}$$

Base Capacitance:

$$C_{base} = \frac{1}{\omega_{base} \cdot Z_{base}} = \frac{1}{314.159 \times 0.13840} = \frac{1}{43.479} = 0.02300 \, \text{F} = 23.00 \, \text{mF}$$

Base Time Constant:

$$\tau_{base} = \frac{1}{\omega_{base}} = \frac{1}{314.159} = 3.183 \, \text{ms}$$

Table 1.1 Complete Per-Unit Base Values for PCS 3440 at 690 V / 3.44 MVA

Quantity	Symbol	Formula	Value	Unit
Base Apparent Power	S_{base}	Nameplate rated	3,440,000	VA
Base Voltage (L-L, RMS)	$V_{base,LL}$	Nameplate rated	690	V
Base Voltage (L-N, RMS)	$V_{base,LN}$	$V_{LL}/\sqrt{3}$	398.37	V
Base Current (3-phase)	I_{base}	$S/(\sqrt{3}V_{LL})$	2,879.0	A
Base Impedance	Z_{base}	V_{LL}^2/S	0.13840	Ω
Base Frequency	f_{base}	Grid nominal	50	Hz
Base Angular Frequency	ω_{base}	$2\pi f$	314.159	rad/s
Base Inductance	L_{base}	Z/ω	0.4404	mH
Base Capacitance	C_{base}	$1/(\omega Z)$	23.00	mF
Base Time Constant	τ_{base}	$1/\omega$	3.183	ms

Per-Unit Conversion Note

Any physical quantity $X_{physical}$ is converted to per-unit as $X_{pu} = X_{physical} / X_{base}$. For example, a filter inductor of $L_f = 0.088$ mH has a per-unit value of $L_{f,pu} = 0.088 / 0.4404 = 0.200$ pu. All control gains in Parts 2–6 are expressed in per-unit unless explicitly stated otherwise.

1.2 PCS 3440 System Ratings and Specifications

The PCS 3440 is a three-phase, voltage-source inverter (VSI) platform rated for grid-scale energy storage and renewable integration. Its architecture supports both grid-following (GFL) and grid-forming (GFM) operation, making it suitable for weak-grid and islanded microgrid environments. Key electrical and control specifications are summarized below.

Table 1.2 PCS 3440 Electrical Specifications

Parameter	Symbol	Value	Unit	Notes
Rated AC Power	P_{rated}	3,440	kW	Unity PF output
Rated Apparent Power	S_{rated}	3,440	kVA	Base for pu system
Rated Reactive Power	Q_{rated}	$\pm 1,720$	kVAr	± 0.5 pu range
AC Voltage (L-L, nom.)	V_{AC}	690	V	$\pm 10\%$ tolerance
AC Current (rated)	I_{AC}	2,879	A	RMS per phase
DC Bus Voltage	V_{DC}	1,200	V	Nominal; range 900–1,400 V
Switching Frequency	f_{sw}	5,000	Hz	IGBT PWM carrier
Output Frequency	f_{out}	50/60	Hz	Selectable
Filter Inductance	L_f	0.088	mH	0.200 pu
Filter Capacitance	C_f	1.150	mF	0.050 pu
Control Loop Bandwidth	f_{BW}	500–1,000	Hz	Inner current loop
Voltage Loop Bandwidth	$f_{BW,V}$	50–100	Hz	Outer voltage loop
Max. Fault Current	$I_{fault,max}$	1.5	pu	Virtual impedance limited
Overload Capacity	I_{OL}	1.1	pu	60-second overload

Efficiency (rated load)	η	>98.0	%	At rated power and voltage
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1.2.1 LC Filter Design Overview

The LC output filter is designed to attenuate switching harmonics while maintaining adequate bandwidth for the voltage control loop. The filter inductance $L_f = 0.200$ pu and capacitance $C_f = 0.050$ pu yield a resonance frequency of:

$$\begin{aligned} f_{res} &= \frac{1}{2\pi\sqrt{L_f C_f}} = \frac{1}{2\pi\sqrt{L_{base} \cdot 0.200 \times C_{base} \cdot 0.050}} \\ &= \frac{1}{2\pi\sqrt{4.404 \times 10^{-4} \times 0.200 \times 0.02300 \times 0.050}} = \frac{1}{2\pi\sqrt{1.013 \times 10^{-7}}} \approx 500 \text{ Hz} \end{aligned}$$

This places the LC resonance frequency at exactly one-tenth of the switching frequency (5 kHz), providing >40 dB of harmonic attenuation at the switching frequency while the filter corner frequency remains well above the voltage control bandwidth of 50–100 Hz.

1.2.2 Grid-Forming vs. Grid-Following Mode Summary

Control Mode Comparison		
Feature	Grid-Following (GFL)	Grid-Forming (GFM)
Grid Synchronization	PLL-based (θ_{PLL})	VSM/Droop self-generated θ
Output Quantity	Current source (I_d^* , I_q^*)	Voltage source (V^* , θ^*)
Frequency Reference	Measured from grid	Self-determined by VSM/Droop
Inertia Contribution	None (zero virtual inertia)	$H = 4$ s (virtual inertia)
Island Mode Capable	No	Yes
Fault Current Limiting	Current limit only	Virtual impedance + current limit

PART 2 — DROOP CONTROL DESIGN

Part 2: Droop Control Design

2.1 P-f Droop Control (4–5% Droop Characteristic)

Active power-frequency (P-f) droop control emulates the governor response of a synchronous generator: when the inverter delivers more active power to the grid, its frequency reference is reduced proportionally. This proportional response allows multiple grid-forming units to share load autonomously without a centralized communication link, following the principles of IEEE Std 1547-2018, Clause 6.5.

2.1.1 Droop Characteristic Definition

The droop percentage $D_P\%$ is defined as the fractional change in frequency at full rated power deviation, expressed as a percentage of nominal frequency:

$$D_P\% = \frac{\Delta f / f_0}{\Delta P / P_{base}} \times 100\%$$

For a droop setting of 5%, the frequency deviation at full power change ($\Delta P = 1$ pu) equals 5% of the nominal frequency. This gives the droop slope m_P in per-unit:

$$m_P = \frac{\Delta \omega_{pu}}{\Delta P_{pu}} = \frac{0.05 \times \omega_{base,pu}}{\Delta P_{pu}} = \frac{0.05 \times 1.0}{1.0} = 0.05 \text{ pu/pu}$$

2.1.2 Transfer Function and Mathematical Model

The P-f droop generates a frequency reference ω^* by subtracting the scaled power error from the nominal angular frequency. A first-order low-pass filter $G_{LP}(s)$ is applied to the measured power to remove switching noise and high-frequency disturbances:

$$G_{LP}(s) = \frac{\omega_f}{s + \omega_f}, \quad \omega_f = 2\pi \times 10 = 62.83 \text{ rad/s}$$

The complete P-f droop transfer function block is:

$$\omega^*(s) = \omega_0 - m_P \cdot G_{LP}(s) \cdot P_{meas}(s) + m_P \cdot P_{ref}(s)$$

In steady-state ($s \rightarrow 0$, $G_{LP}(0) = 1$):

$$\omega_{ss}^* = \omega_0 - m_P(P_{meas} - P_{ref}) = \omega_0 - m_P \cdot \Delta P$$

The droop equation expressed in per-unit quantities (all quantities normalized to base values):

$$\omega_{pu}^* = 1.0 - m_{P,pu} \cdot (P_{meas,pu} - P_{ref,pu})$$

P-f Droop Design Equations

For 4% droop: $m_P = 0.04$ pu, frequency range $\pm 2\%$ at full load swing.

For 5% droop: $m_P = 0.05$ pu, frequency range $\pm 2.5\%$ at full load swing.

Recommended setting for PCS 3440 islanded operation: $m_P = 0.05$ pu (5% droop).

Power filter bandwidth: $\omega_f = 62.83$ rad/s (10 Hz low-pass filter).

2.1.3 Droop Coefficient Calculation in Per-Unit

Starting from the physical droop slope in Hz/MW:

$$k_P^{phys} = \frac{\Delta f}{\Delta P} = \frac{0.05 \times 50 \text{ Hz}}{3.44 \text{ MW}} = \frac{2.5}{3.44} = 0.7267 \text{ Hz/MW}$$

Converting to per-unit (pu angular frequency per pu power):

$$m_{P,pu} = k_P^{phys} \times \frac{P_{base}}{f_{base}} = 0.7267 \times \frac{3,440,000}{50} = 0.7267 \times 68,800 = 50,000 \text{ [rad/s/MV]}$$

Normalizing to per-unit on both axes:

$$m_{P,pu} = \frac{\Delta\omega/\omega_{base}}{\Delta P/P_{base}} = \frac{0.05 \times 1.0}{1.0} = 0.05 \text{ pu}$$

The per-unit droop coefficient can also be expressed as an equivalent gain block value in PSCAD:

$$K_{droop,P} = m_P = 0.05 \text{ pu/pu} \Rightarrow \text{Gain block value} = 0.05$$

2.1.4 Frequency vs. Active Power Characteristic Curve

The P-f characteristic is a straight line with slope $m_P = 0.05$ pu, operating between two characteristic points:

Table 2.1 P-f Droop Operating Points (5% Droop, 50 Hz Nominal)

Operating Point	P (pu)	P (kW)	ω^* (pu)	f* (Hz)	Δf (Hz)
No-load (P=0)	0.00	0	1.050	52.50	+2.50
25% load	0.25	860	1.0375	51.875	+1.875
50% load (reference)	0.50	1,720	1.025	51.25	+1.25
Rated load (P=1.0 pu)	1.00	3,440	1.000	50.00	0

Overload (P=1.1 pu)	1.10	3,784	0.9950	49.75	-0.25
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2.1.5 PSCAD Component Blocks for P-f Droop

PSCAD Implementation — P-f Droop Block

The P-f droop is implemented in the PSCAD *Controls Library* using the following sequence of component blocks:

1. **Three-Phase Power Meter** (Library: Meters > Power): Measures three-phase instantaneous real power P_{meas} . Set base: 3.44 MVA. Output: P in per-unit.
2. **First-Order Low-Pass Filter** (Library: CSMF > Filters > First Order): Time constant $T_f = 1/\omega_f = 15.92$ ms. Apply to P_{meas} signal.
3. **Summing Junction** (Library: CSMF > Arithmetic): Compute $\Delta P = P_{meas,filtered} - P_{ref}$. $P_{ref} = 1.0$ pu from external reference.
4. **Gain Block** (Library: CSMF > Gain): Value = 0.05 (= m_P). Multiplies ΔP by droop coefficient.
5. **Negation + Summing Junction**: Compute $\omega^* = \omega_0 - m_P \cdot \Delta P = 1.0 - 0.05 \cdot \Delta P$.
6. **Limiter Block** (Library: CSMF > Limiters > Hard Limiter): Clamp ω^* between 0.95 pu and 1.05 pu (47.5–52.5 Hz) per IEEE 1547-2018.

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Component:First-Order LP Filter (Pf Droop)
  Time Constant T = 0.01592 s (= 1/62.83 rad/s)
  Initial Value = 1.0 pu
Component:Gain Block (Kp_droop)
  Gain = 0.05 pu/pu
Component:Hard Limiter (omega_ref)
  Upper Limit = 1.05 pu   Lower Limit = 0.95 pu

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2.2 Q-V Droop Control (5% Droop Characteristic)

Reactive power-voltage (Q-V) droop control provides local voltage regulation by adjusting the terminal voltage reference proportionally to the reactive power output. Unlike P-f droop (which is nearly universal across generator types), Q-V droop is more system-dependent because reactive power flow is strongly influenced by network impedances and proximity of generation/load.

2.2.1 Q-V Droop Mathematical Derivation

The Q-V droop characteristic is defined as:

$$V^*(pu) = V_0 - m_Q \cdot Q_{meas,pu}$$

Where m_Q is the Q-V droop coefficient (pu voltage drop per pu reactive power). For 5% droop at rated reactive power ($Q_{rated} = 0.5$ pu for unity-PF-capable converter), the voltage drops 5% :

$$m_Q = \frac{\Delta V_{pu}}{\Delta Q_{pu}} = \frac{0.05}{1.0} = 0.05 \text{ pu/pu}$$

However, since the PCS 3440 is rated at $Q_{rated,pu} = 0.5$ pu (1720 kVAr), a 5% voltage droop at rated Q requires:

$$m_Q = \frac{0.05 \text{ pu (voltage)}}{0.5 \text{ pu (reactive power)}} = 0.10 \text{ pu/pu}$$

The complete Q-V droop transfer function with first-order reactive power filter:

$$V^*(s) = V_0 + m_Q \cdot (Q_{ref} - G_{LP,Q}(s) \cdot Q_{meas}(s))$$

$$G_{LP,Q}(s) = \frac{\omega_{fQ}}{s + \omega_{fQ}}, \quad \omega_{fQ} = 2\pi \times 5 = 31.42 \text{ rad/s}$$

The reactive power filter uses a lower bandwidth (5 Hz) compared to the active power filter (10 Hz) to improve voltage stability in island mode, where reactive power dynamics are slower.

2.2.2 Q-V Droop Characteristic Operating Points

Table 2.2 Q-V Droop Operating Points (5% at Q=0.5 pu, V0=1.0 pu)

Q (pu)	Q (kVAr)	Type	V* (pu)	V* (V, L-L)
-0.50	-1,720	Full capacitive (absorbing Q)	1.050	724.5
-0.25	-860	Partial capacitive	1.025	707.3
0.00	0	Unity power factor	1.000	690.0
+0.25	+860	Partial inductive (injecting Q)	0.975	672.8
+0.50	+1,720	Full inductive	0.950	655.5

2.2.3 PSCAD Implementation of Q-V Droop

PSCAD Implementation — Q-V Droop Block

1. **Reactive Power Meter** (Library: Meters > Power): Output Q_{meas} in per-unit (base 3.44 MVA).
2. **First-Order LP Filter (Q)**: Time constant $T_{fQ} = 1/31.42 = 31.83$ ms. Filters reactive power measurement.
3. **Error Computation** (Summing Junction): $\Delta Q = Q_{ref} - Q_{meas,filtered}$. Set $Q_{ref} = 0$ for unity-PF reference.
4. **Gain Block (mQ)**: Value = 0.10 pu/pu. Scales reactive power error to voltage deviation.
5. **Voltage Reference Summation**: $V^* = V_0 + m_Q \cdot \Delta Q = 1.0 + 0.10 \cdot \Delta Q$.
6. **Hard Limiter (V_ref)**: Clamp to [0.88, 1.10] pu per IEEE 1547-2018 voltage regulation range.

Component:First-Order LP Filter (QV Droop)
Time Constant T = 0.03183 s (= 1/31.42 rad/s)
Component:Gain Block (Kq_droop)
Gain = 0.10 pu/pu
Component:Hard Limiter (V_ref)
Upper Limit = 1.10 pu Lower Limit = 0.88 pu

2.2.4 Summary of Droop Parameters

Table 2.3 Summary of PCS 3440 Droop Control Parameters

Parameter	Symbol	Value (pu)	Physical Value
P-f Droop Coefficient	m_P	0.05	0.7267 Hz/MW
P-f Filter Time Constant	T_{fP}	5.00 pu	15.92 ms
P-f Frequency Limits	$\omega_{min}/\omega_{max}$	0.95 / 1.05	47.5 / 52.5 Hz
Q-V Droop Coefficient	m_Q	0.10	0.0291 kV/kVAr
Q-V Filter Time Constant	T_{fQ}	10.0 pu	31.83 ms
Q-V Voltage Limits	V_{min}/V_{max}	0.88 / 1.10	607.2 / 759.0 V
Nominal Frequency	ω_0	1.000	50 Hz / 314.159 rad/s
Nominal Voltage	V_0	1.000	690 V (L-L)

PART 3 — VIRTUAL SYNCHRONOUS MACHINE (VSM) DESIGN

Part 3: Virtual Synchronous Machine (VSM) Design

3.1 VSM Mathematical Model and Swing Equation

The Virtual Synchronous Machine (VSM) is a control algorithm that impresses the electromechanical behavior of a synchronous generator onto a power electronic inverter. By emulating the rotor swing equation, the VSM provides frequency-responsive inertia, damping, and self-synchronization properties that a conventional current-controlled inverter completely lacks. The VSM approach is formalized in IEEE P2800.1 and documented in IEC TR 63227 for inverter-based resources.

3.1.1 Swing Equation of a Synchronous Machine

The classical swing equation for a synchronous generator describes the rotational dynamics of the rotor mass:

$$\frac{2H}{\omega_0} \frac{d\omega}{dt} = T_m - T_e - D(\omega - \omega_0)$$

Dividing through by the base torque $T_{base} = S_{base}/\omega_0$ and expressing in per-unit angular frequency $\omega_{pu} = \omega/\omega_0$:

$$2H \frac{d\omega_{pu}}{dt} = P_m - P_e - D(\omega_{pu} - 1)$$

Where:

- H = Inertia constant [seconds] — virtual inertia for VSM
- ω_{pu} = Per-unit angular frequency ($= \omega/\omega_{base}$)
- P_m = Virtual mechanical power input [pu] — set by droop control
- P_e = Electrical power delivered to grid [pu] — measured
- D = Damping coefficient [pu torque / pu speed] — virtual damping

Solving for the angular frequency deviation:

$$\frac{d\omega_{pu}}{dt} = \frac{1}{2H} [P_m - P_e - D(\omega_{pu} - 1)]$$

Integrating to obtain the virtual rotor angle θ :

$$\frac{d\theta}{dt} = \omega = \omega_0 \cdot \omega_{pu}(t)$$

3.1.2 Transfer Function of the VSM Governor Block

Taking the Laplace transform of the swing equation and rearranging to express $\omega_{pu}(s)$ as a function of power imbalance $\Delta P(s) = P_m(s) - P_e(s)$:

$$\omega_{pu}(s) = \frac{1}{2Hs + D} \cdot \Delta P(s) + \frac{D}{2Hs + D}$$

The closed-loop transfer function from power error to frequency (damped second-order response):

$$G_{VSM}(s) = \frac{\omega_{pu}(s)}{\Delta P(s)} = \frac{1}{2Hs + D}$$

The characteristic time constant of the VSM governor dynamics:

$$\tau_{VSM} = \frac{2H}{D} = \frac{2 \times 4}{20} = 0.40 \text{ s} = 400 \text{ ms}$$

The VSM governor bandwidth (natural frequency of frequency response):

$$\omega_{VSM} = \frac{D}{2H} = \frac{20}{8} = 2.5 \text{ rad/s} \approx 0.40 \text{ Hz}$$

3.2 Virtual Inertia ($H = 4 \text{ s}$) and Damping ($D = 20$)

3.2.1 Inertia Constant H — Physical Interpretation

The inertia constant H is the ratio of the kinetic energy stored in the rotating mass at synchronous speed to the machine's rated apparent power:

$$H = \frac{E_{kinetic}}{S_{base}} = \frac{\frac{1}{2} J \omega_0^2}{S_{base}} \quad [\text{seconds}]$$

For the VSM, there is no physical spinning mass. Instead, H is a control parameter that determines how rapidly the virtual rotor speed responds to power imbalances. A higher H provides more inertial support (slower frequency changes) but slower dynamic response to set-point changes.

For PCS 3440 with $H = 4 \text{ s}$:

$$J_{virtual} = \frac{2H \cdot S_{base}}{\omega_0^2} = \frac{2 \times 4 \times 3,440,000}{(314.159)^2} = \frac{27,520,000}{98,715.9} = 278.76 \text{ kg} \cdot \text{m}^2$$

In per-unit, the inertia coefficient $M = 2H = 8 \text{ pu} \cdot \text{s}$, which is the time in seconds taken to accelerate the virtual rotor from standstill to synchronous speed under rated torque (time constant of frequency response).

3.2.2 Damping Coefficient D — Physical Interpretation

The damping coefficient D represents the friction and load-damping effects of a synchronous machine. In the VSM context, D provides active power injection proportional to frequency deviation, acting as a frequency-responsive damper:

$$\Delta P_{damping} = D \cdot (\omega_{pu} - 1.0)$$

At a frequency deviation of $\Delta\omega = 0.01$ pu (0.5 Hz from 50 Hz nominal), the damping power is:

$$\Delta P_{damp} = 20 \times 0.01 = 0.20 \text{ pu} = 0.20 \times 3,440 = 688 \text{ kW}$$

This 688 kW of damping injection at a 0.5 Hz deviation is significant — approximately 20% of rated power. The $D = 20$ setting provides strong frequency stabilization without over-damping the VSM response.

3.2.3 Per-Unit Parameter Summary

Table 3.1 VSM Parameters in Per-Unit for PCS 3440

Parameter	Symbol	Value	Unit	Physical Equivalent
Inertia Constant	H	4.000	s	278.76 kg·m ² virtual inertia
2H (inertia coefficient)	M	8.000	pu·s	Integrator gain in swing eq.
Damping Coefficient	D	20.00	pu/pu	688 kW/0.5 Hz deviation
VSM Time Constant	τ_{VSM}	0.400	s	Governor response time
VSM Natural Frequency	ω_{VSM}	2.500	rad/s	0.398 Hz oscillation freq.
Virtual Rotor Inertia	J	278.76	kg·m ²	Equivalent physical mass
Nominal Angular Freq.	ω_0	1.000	pu	314.159 rad/s (50 Hz)

3.3 PSCAD Implementation of VSM

The VSM swing equation is implemented in PSCAD using standard CSMF (Continuous System Model Functions) components. The implementation follows a modular structure with clearly labeled signal nodes that match the mathematical formulation.

3.3.1 Swing Equation PSCAD Block Chain

PSCAD VSM Swing Equation Implementation

Signal flow from power inputs to angle output:

1. **Droop Power Reference P_m** : Input from P-f droop block. In per-unit, this is the virtual mechanical power setpoint including the droop correction: $P_m = P_{ref} - m_P \cdot \Delta f$.
2. **Measured Electrical Power P_e** : From Three-Phase Power Meter block. Low-pass filtered with $T_f = 10$ ms to remove switching noise.
3. **Power Error Summation** (Summing Junction, gains: +1 for P_m , -1 for P_e): Computes $\Delta P = P_m - P_e$.
4. **Damping Term Subtraction** (Summing Junction): Computes $\Delta P_{net} = \Delta P - D \cdot (\omega_{pu} - 1)$. The frequency deviation signal $(\omega_{pu} - 1)$ is fed back from the integrator output.
5. **Inertia Gain Block**: Gain = $1/(2H) = 1/8 = 0.125$ pu. Converts net power to angular acceleration $d\omega/dt$.
6. **Integrator Block #1** (Library: CSMF > Integrator): Input = $d\omega_{pu}/dt$. Initial value = 1.0 pu. Output = $\omega_{pu}(t)$. Limits: [0.95, 1.05] pu.
7. **Multiplier**: $\omega_{phys} = \omega_{pu} \times \omega_0 = \omega_{pu} \times 314.159$ rad/s.
8. **Integrator Block #2** (Library: CSMF > Integrator): Input = ω_{phys} . Output = $\theta(t)$ [radians]. Initial value = 0. No limits (free-running angle).
9. **Modulo Block** (Library: CSMF > Math): Computes $\theta_{mod} = \theta \bmod 2\pi$ to keep angle bounded. Feed to SPWM modulator.

```

Component:Gain (1/2H)      Gain = 0.125 pu (= 1/8)
Component:Gain (D_vsm)    Gain = 20.0 pu
Component:Integrator (omega)  IC = 1.0 pu   Limits: [0.95, 1.05] pu
Component:Gain (omega_base) Gain = 314.159 rad/s
Component:Integrator (theta) IC = 0.0 rad   No limits

```

3.3.2 VSM Feedback Loop and Signal Routing

A critical aspect of the PSCAD VSM implementation is the feedback of the frequency signal ω_{pu} back to the damping subtraction junction. This creates an algebraic loop in the block diagram, which must be broken with a small delay (one simulation time-step, typically 50 μ s) to avoid instantaneous circular dependency. In PSCAD V5.0.2, this is handled automatically by the EMTDC solver when the signal path is correctly wired through the integrator output.

Algebraic Loop Warning — PSCAD Implementation

Do NOT connect the damping feedback $D \cdot (\omega_{pu} - 1)$ directly from a non-integrator node back to an input of the same computation. Always route through the Integrator #1 output. In PSCAD V5.0.2, right-click the integrator and verify "Output Delay = 1 time-step" is enabled. This adds one $\Delta t = 50 \mu s$ delay to the feedback path — negligible for the 400 ms VSM dynamics but essential for numerical stability.

3.3.3 VSM Output Signal Processing

The VSM block produces two key outputs used downstream by the modulator and voltage/current control loops:

- **Virtual rotor angle θ_{VSM} :** Used as the reference angle for Park (dq) transformation of voltages and currents, and as the phase angle argument for the SPWM sinusoidal reference: $v_{ref,a} = V^* \cdot \sqrt{2} \cdot \sin(\theta_{VSM})$.
- **Virtual rotor frequency ω_{VSM} :** Fed back to the droop controller for frequency-responsive power sharing, to the ROCOF detection block, and to the seamless transition logic (Part 6).

PART 4 — VOLTAGE AND FREQUENCY REGULATION (ISLAND MODE)**Part 4: Voltage and Frequency Regulation (Island Mode)****4.1 Voltage Control Loop Design**

In grid-forming (island) mode, the PCS 3440 must regulate the load bus voltage with high precision. Unlike grid-connected mode where the grid stiffens the voltage, island mode requires the inverter to establish and maintain the voltage reference autonomously. A cascaded control structure is employed: an outer voltage controller sets the current reference for an inner current controller, which directly commands the PWM modulator.

4.1.1 Cascaded Voltage-Current Control Architecture

The cascaded structure in the dq-reference frame (rotating at ω_{VSM} from the VSM block) consists of:

- **Outer loop:** Voltage PI controller — tracks capacitor voltage $V_{C,dq}$ to reference V_{dq}^* from droop/VSM, bandwidth $f_{BW,V} = 75 \text{ Hz}$.

- **Inner loop:** Current PI controller — tracks inductor current $I_{L,dq}$ to reference I_{dq}^* from outer loop, bandwidth $f_{BW,I} = 750$ Hz.
- **Feed-forward:** Capacitor current feed-forward reduces outer-loop integral action and improves transient response.

4.1.2 Voltage Controller PI Design

The voltage controller operates on the capacitor voltage error in the dq frame. The design target is a closed-loop bandwidth of 75 Hz (470 rad/s), with a phase margin of 60°. The plant seen by the voltage controller (assuming ideal inner current loop) is an integrator through the capacitor:

$$G_{plant,V}(s) = \frac{1}{sC_f} = \frac{1}{s \times 1.150 \times 10^{-3}} = \frac{869.6}{s}$$

In per-unit ($C_{f,pu} = 0.050$ pu, $\tau_{base} = 3.183$ ms):

$$G_{plant,V,pu}(s) = \frac{1}{sC_{f,pu}} = \frac{1}{s \times 0.050} = \frac{20.0}{s} \text{ pu}$$

For a PI controller $C_V(s) = K_{pV}(1 + 1/(sT_{iV}))$ with the zero at one-tenth the crossover frequency to provide adequate phase margin:

$$T_{iV} = \frac{10}{\omega_{BW,V}} = \frac{10}{2\pi \times 75} = \frac{10}{471} = 21.23 \text{ ms}$$

Setting the proportional gain to achieve $\omega_{BW,V} = 471$ rad/s crossover with the integrator plant:

$$K_{pV} = \frac{\omega_{BW,V} \times C_{f,pu}}{1} = 471 \times 0.050 = 23.55 \text{ pu}$$

The integral gain:

$$K_{iV} = \frac{K_{pV}}{T_{iV}} = \frac{23.55}{0.02123} = 1,110 \text{ pu/s}$$

Alternatively, expressed as $K_{iV} = K_{pV} \cdot \omega_{z,V}$ where $\omega_{z,V} = 1/T_{iV} = 47.1$ rad/s:

$$K_{iV} = 23.55 \times 47.1 = 1,109 \text{ pu/s}$$

Table 4.1 Voltage PI Controller Parameters

Parameter	Symbol	Value (pu)	Physical Value	Notes
Proportional Gain	K_{pV}	23.55	23.55 A/V	d-axis and q-axis (identical)
Integral Gain	K_{iV}	1,109	1,109 A/(V·s)	Ensures zero steady-state error

Integrator Time Const.	T_{iV}	6.667 pu	21.23 ms	PI zero at 47.1 rad/s
Controller Bandwidth	$f_{BW,V}$	—	75 Hz (471 rad/s)	Closed-loop with inner loop ideal
Phase Margin	ϕ_m	—	$\sim 60^\circ$	With zero at 47.1 rad/s
Output Limiter	I_{limit}^*	1.10 pu	3,167 A	Current reference clamped

4.1.3 Voltage Reference Generation

The voltage reference in the dq frame is generated from the VSM angle θ_{VSM} and the droop voltage magnitude V^* :

$$V_d^* = V^* \cos(0) = V^* \quad [\text{d-axis aligned with terminal voltage}]$$

$$V_q^* = V^* \sin(0) = 0 \quad [\text{q-axis voltage reference} = 0 \text{ for unity PF reference}]$$

The magnitude $V^* = 1.0$ pu (adjusted by Q-V droop) and the angle θ_{VSM} from the VSM block together define the complete voltage reference vector. In PSCAD, this is implemented using a Controlled Voltage Source driven by:

$$v_{ref,a}(t) = V^* \cdot \sqrt{2} \cdot \sin(\theta_{VSM}(t))$$

$$v_{ref,b}(t) = V^* \cdot \sqrt{2} \cdot \sin(\theta_{VSM}(t) - 2\pi/3)$$

$$v_{ref,c}(t) = V^* \cdot \sqrt{2} \cdot \sin(\theta_{VSM}(t) + 2\pi/3)$$

4.1.4 PSCAD Components for Voltage Control Loop

PSCAD Voltage Controller (Outer Loop) Components

1. **3-phase Voltage Measurement** (Library: Meters > Voltage): Measure capacitor voltage $V_{C,abc}$. Use RMS measurement for magnitude and dq-decomposition.
2. **Park Transform (abc → dq)** (Library: CSMF > Transforms > 3ph to dq): Angle input = θ_{VSM} . Outputs: $V_{C,d}$, $V_{C,q}$.
3. **Voltage PI (d-axis)** (Library: CSMF > Controllers > PI): Kp=23.55, Ki=1109. Input: $V_d^* - V_{C,d}$. Output: I_d^* .
4. **Voltage PI (q-axis)** (Library: CSMF > Controllers > PI): Kp=23.55, Ki=1109. Input: $V_q^* - V_{C,q} = 0 - V_{C,q}$. Output: I_q^* .
5. **Anti-Windup Limiter**: Current reference $I_{d,q}^*$ clamped to ± 1.10 pu with integrator back-calculation.
6. **Feed-Forward Block**: Add $I_{C,d}$ (capacitor current) to output of voltage PI to decouple capacitor current disturbance.

Component:PI Controller (Voltage d-axis)

Kp = 23.55 Ki = 1109 Output limit = ± 1.10 pu

Component:PI Controller (Voltage q-axis)

Kp = 23.55 Ki = 1109 Output limit = ± 1.10 pu

4.2 Frequency Control Loop Design

In island mode, frequency is not imposed externally — it is generated by the VSM integrator chain. The frequency control loop monitors the VSM output frequency and, if necessary, provides a corrective signal to the virtual mechanical power input P_m to restore frequency to the nominal value. The primary frequency response is provided by the droop control (Part 2), while the VSM inertia provides the short-term buffering.

4.2.1 Frequency Measurement and Filtering

The VSM outputs ω_{pu} directly. For external frequency measurement (used in islanding detection, Part 6), a zero-crossing phase detector or a second-order generalized integrator (SOGI) is employed:

$$G_{SOGI}(s) = \frac{k\omega's}{s^2 + k\omega's + \omega'^2}$$

Where $k = \sqrt{2}$ (standard SOGI gain) and ω' is the estimated frequency. The SOGI output provides a filtered fundamental-frequency component, from which frequency is derived by tracking the zero-crossings. In PSCAD V5.0.2, use:

- Library: Custom CSMF Subsystem or built-in PLL block (Library: Machine/Control > PLL > DSOGI-PLL)
- Loop filter bandwidth: 30 Hz for fast tracking, or 10 Hz for noise rejection

4.2.2 Secondary Frequency PI Controller

A slow outer PI controller restores frequency to nominal after primary droop response, emulating the AGC (Automatic Generation Control) function. This is optional in island mode but critical for frequency quality:

$$C_f(s) = K_{pf} + \frac{K_{if}}{s}, \quad K_{pf} = 0.5 \text{ pu}, \quad K_{if} = 0.2 \text{ pu/s}$$

The secondary frequency controller acts on the integral of frequency error and modifies P_{ref} over a 5–30 second timescale. Its output is clamped to ± 0.2 pu to limit its impact on the primary droop response.

Table 4.2 Frequency Control PI Parameters (Island Mode)

Parameter	Symbol	Value	Notes
Proportional Gain	K_{pf}	0.50 pu	Primary droop dominates; Kpf is supplementary
Integral Gain	K_{if}	0.20 pu/s	Slow AGC action, 5–30 s timescale
Output Limit	$\Delta P_{AGC,max}$	± 0.20 pu	686 kW maximum AGC correction
Frequency Deadband	f_{DB}	± 0.01 Hz	No AGC action inside deadband
AGC Activation Delay	T_{AGC}	2.0 s	Allows primary response to settle first

4.2.3 Inner Current Control Loop

The inner current PI controller tracks the current reference generated by the outer voltage controller. Its bandwidth must be approximately 10× the outer voltage loop bandwidth:

$$f_{BW,I} = 10 \times f_{BW,V} = 10 \times 75 = 750 \text{ Hz}$$

The plant for the current controller is the filter inductance $L_f = 0.200$ pu in series with parasitic resistance $R_f = 0.01$ pu:

$$G_{plant,I}(s) = \frac{1}{sL_f + R_f} = \frac{1}{s \times 0.200 + 0.01} = \frac{5.0}{s + 0.05}$$

Using PI controller with zero cancellation of the plant pole:

$$T_{iI} = \frac{L_f}{R_f} = \frac{0.200}{0.01} = 20.0 \text{ pu} \cdot \text{s} \quad (\text{physical: } 63.66 \text{ ms})$$

$$K_{pI} = \omega_{BW,I} \times L_f = (2\pi \times 750) \times 0.200 = 4,712 \times 0.200 = 942.5 \text{ pu}$$

$$K_{iI} = K_{pI} \times R_f / L_f = 942.5 \times 0.05 = 47.12 \text{ pu/s}$$

Component:PI Controller (Current d-axis)

Kp = 942.5 Ki = 47.12 Output limit = ± 1.15 pu (modulator index limit)

Component:PI Controller (Current q-axis)

Kp = 942.5 Ki = 47.12 Output limit = ± 1.15 pu

Component:Cross-coupling Decoupler: ωL_f feed-forward to eliminate dq-axis coupling

Gain = 0.200 pu (= $L_{f,pu}$)

PART 5 — VIRTUAL IMPEDANCE DESIGN

Part 5: Virtual Impedance Design

5.1 Mathematical Derivation and Per-Unit Values

Virtual impedance is a software-implemented impedance that modifies the inverter voltage reference to create the effect of a physical series impedance at the inverter output, without the losses or physical components associated with a real impedance. It serves three critical functions in grid-forming operation: (1) fault current limitation, (2) harmonic current suppression, and (3) power sharing improvement in microgrid parallel operation.

5.1.1 Virtual Impedance Concept

A physical series impedance $Z_{virt} = R_{virt} + j\omega L_{virt}$ placed between the inverter and the LC filter capacitor would produce a voltage drop:

$$\Delta V_{phys} = Z_{virt} \cdot I_L = (R_{virt} + j\omega L_{virt}) \cdot I_L$$

Virtual impedance replicates this drop in the control domain by subtracting an equivalent voltage compensation from the inverter voltage reference:

$$V_{inv,modified}^* = V_{inv,original}^* - \Delta V_{virt}$$

$$\Delta V_{virt,d} = R_{virt} \cdot I_{L,d} - \omega L_{virt} \cdot I_{L,q}$$

$$\Delta V_{virt,q} = R_{virt} \cdot I_{L,q} + \omega L_{virt} \cdot I_{L,d}$$

The complete virtual impedance voltage compensation in dq-frame (including the cross-coupling terms from the angular frequency rotation):

$$\begin{bmatrix} \Delta V_{virt,d} \\ \Delta V_{virt,q} \end{bmatrix} = \begin{bmatrix} R_{virt} & -\omega L_{virt} \\ \omega L_{virt} & R_{virt} \end{bmatrix} \begin{bmatrix} I_{L,d} \\ I_{L,q} \end{bmatrix}$$

5.1.2 Fault Current Limiting Logic

Under normal operation ($|I_L| \leq 1.1$ pu), the virtual impedance is set to zero (or a small background value for harmonic sharing). When overcurrent is detected ($|I_L| > 1.1$ pu), the virtual impedance is activated and scaled to limit the fault current to 1.5 pu maximum:

$$|I_{L,max}| = 1.5 \text{ pu} \Rightarrow I_{fault,limit} = 1.5 \times 2,879 = 4,318.5 \text{ A}$$

The required virtual impedance magnitude to limit current from $|I_{fault,unlimited}|$ to $|I_{fault,limit}| = 1.5$ pu, given the pre-fault voltage $|V_0| = 1.0$ pu:

$$|Z_{virt}| = \frac{|V_{fault}|}{|I_{fault,limit}|} - |Z_{grid}|$$

For a bolted three-phase fault at the PCC with $|Z_{grid}| = 0.05$ pu (stiff grid assumption):

$$|Z_{virt}| = \frac{1.0}{1.5} - 0.05 = 0.6667 - 0.05 = 0.617 \text{ pu}$$

5.1.3 R_virtual and L_virtual Calculations

The virtual impedance is designed as predominantly inductive (high X/R ratio = 10) to avoid active power coupling effects during faults. Choosing X/R = 10:

$$\tan(\phi_{virt}) = X_{virt}/R_{virt} = 10 \Rightarrow \phi_{virt} = \arctan(10) = 84.3^\circ$$

$$|Z_{virt}| = \sqrt{R_{virt}^2 + X_{virt}^2} = R_{virt}\sqrt{1 + 100} = R_{virt} \times 10.05$$

Solving for R_{virt} and $X_{virt} = \omega L_{virt}$:

$$R_{virt} = \frac{|Z_{virt}|}{10.05} = \frac{0.617}{10.05} = 0.0614 \text{ pu} = 0.0614 \times 0.13840 = 0.00850 \Omega$$

$$X_{virt} = 10 \times R_{virt} = 10 \times 0.0614 = 0.614 \text{ pu}$$

$$L_{virt} = \frac{X_{virt}}{\omega_0} = \frac{0.614}{1.0} = 0.614 \text{ pu} = 0.614 \times 0.4404 \text{ mH} = 0.2704 \text{ mH}$$

5.1.4 Scaled Virtual Impedance with Current Magnitude

The virtual impedance is activated through a smooth saturation function rather than a hard switch to avoid abrupt transients. The activation gain k_{virt} is a function of the per-unit current magnitude $|I|_{pu}$:

$$k_{virt}(|I|_{pu}) = \begin{cases} 0 & \text{if } |I|_{pu} \leq 1.0 \\ \frac{|I|_{pu}-1.0}{0.5} & \text{if } 1.0 < |I|_{pu} \leq 1.5 \\ 1.0 & \text{if } |I|_{pu} > 1.5 \end{cases}$$

The effective virtual impedance applied is $Z_{eff} = k_{virt} \times Z_{virt,max}$. This linear ramp between 1.0 and 1.5 pu current provides smooth impedance injection without abrupt jumps in the voltage reference.

Table 5.1 Virtual Impedance Parameters for PCS 3440

Parameter	Symbol	Value (pu)	Physical Value
Virtual Resistance (max)	$R_{virt,max}$	0.0614	8.50 mΩ
Virtual Reactance (max)	$X_{virt,max}$	0.614	85.0 mΩ
Virtual Inductance (max)	$L_{virt,max}$	0.614	0.2704 mH
Virtual Impedance Magnitude	$ Z_{virt} $	0.617	85.4 mΩ
X/R Ratio	X/R	10.0	—
Activation Threshold	I_{act}	1.10	3,167 A
Max. Fault Current Limit	I_{limit}	1.50	4,319 A
Ramp Range	ΔI_{ramp}	0.40	1,152 A

5.2 PSCAD Implementation and Fault Current Limiting

5.2.1 Virtual Impedance PSCAD Block Structure

PSCAD Virtual Impedance Implementation

1. **Current Measurement** (Library: Meters > Current): Measure 3-phase inductor currents I_{La} , I_{Lb} , I_{Lc} .
2. **Park Transform (abc → dq)**: Convert to $I_{L,d}$ and $I_{L,q}$ using θ_{VSM} angle.
3. **Current Magnitude Calculation** (CSMF > Math > Sqrt): $|I| = \sqrt{I_{L,d}^2 + I_{L,q}^2}$.
4. **Activation Gain Calculator** (CSMF > Comparator + Math): Implements the piecewise linear $k_{virt}(|I|)$ function using comparators and arithmetic blocks:
 - Comparator 1: Check $|I| > 1.0$ pu → output 1 if true, 0 if false.
 - Comparator 2: Check $|I| > 1.5$ pu → output 1 if true, 0 if false.
 - Linear interpolation: $k_{virt} = \text{clip}((|I| - 1.0)/0.5, 0, 1)$
5. **R_virt Gain Blocks**: Multiply k_{virt} by $R_{virt,max} = 0.0614$ pu. Apply to both d- and q-axis.
6. **L_virt Gain Blocks**: Multiply k_{virt} by $L_{virt,max} = 0.614$ pu. Apply to cross-coupling terms.
7. **Voltage Compensation Computation**:
 - $\Delta V_d = R_{eff} \cdot I_{L,d} - \omega \cdot L_{eff} \cdot I_{L,q}$
 - $\Delta V_q = R_{eff} \cdot I_{L,q} + \omega \cdot L_{eff} \cdot I_{L,d}$
8. **Subtraction from Voltage Reference** (Summing Junction): $V_{d,mod}^* = V_d^* - \Delta V_d$, $V_{q,mod}^* = V_q^* - \Delta V_q$.

```

Component:Gain (R_virt_max)    Gain = 0.0614 pu
Component:Gain (L_virt_max)    Gain = 0.614 pu
Component:Gain (omega for Lvirt)  Gain = 1.0 pu (=  $\omega/\omega_{base}$ )
Component:Comparator (I_act threshold)  Threshold = 1.10 pu
Component:Comparator (I_limit threshold)  Threshold = 1.50 pu

```

5.2.2 Fault Current Limitation Characteristic

Table 5.2 Virtual Impedance Activation vs. Current Level

Current Level (pu)	k_virt	R_eff (pu)	L_eff (pu)	Max Fault Current at V=1.0 pu
0.00 – 1.10	0.000	0.000	0.000	Unlimited (filter impedance only)
1.20	0.200	0.0123	0.123	1.20 pu (activation just starting)

1.30	0.400	0.0246	0.246	1.30 pu
1.40	0.600	0.0368	0.368	1.40 pu
1.50	1.000	0.0614	0.614	1.50 pu (fully limited)
>1.50	1.000	0.0614	0.614	1.50 pu (clamped)

Important: Harmonic Virtual Impedance

For harmonic current sharing between parallel inverters in a microgrid, a frequency-dependent virtual impedance is added at harmonic frequencies (5th, 7th, 11th, 13th). In PSCAD, this is implemented as a bandpass filter centered at each harmonic frequency in parallel with the fundamental virtual impedance. Typical harmonic virtual resistance: $R_h = 0.02$ pu at 5th and 7th harmonics. This is outside the fault-limiting scope of this document but should be added for multi-unit microgrid applications.

PART 6 — SEAMLESS TRANSITION LOGIC

Part 6: Seamless Transition Logic

6.1 Islanding Detection and Mode Logic

Seamless mode transition is one of the most demanding requirements for a grid-forming inverter. The system must transition between grid-connected (GFL) and island (GFM) modes without significant voltage or current disturbances to the load. The transition algorithm consists of three stages: islanding detection, state machine decision logic, and bumpless transfer execution.

6.1.1 Grid-Following Mode Control Structure

In grid-following (GFL) mode, the PCS 3440 operates as a current-controlled voltage source inverter synchronized to the grid via a Phase-Locked Loop (PLL). The control structure is:

- **Synchronization:** DSOGI-PLL (Dual Second-Order Generalized Integrator PLL) locks to grid voltage at PCC. Output: θ_{PLL} , ω_{PLL} .
- **dq Current Control:** Inner PI current controllers regulate I_d (active power) and I_q (reactive power) directly.
- **Power References:** P_{ref} and Q_{ref} from energy management system (EMS) translate to I_d^* and I_q^* .
- **No Voltage Loop:** Grid establishes voltage; inverter follows.

6.1.2 Grid-Forming Mode Control Structure

In grid-forming (GFM) mode, the PCS 3440 establishes voltage and frequency autonomously:

- **VSM Block:** Generates θ_{VSM} and ω_{VSM} from swing equation (Part 3).
- **Voltage Control:** Cascaded V-I PI controllers regulate terminal voltage to V^* (Part 4).
- **Droop Control:** P-f and Q-V droop adjust ω_{VSM} and V^* for load sharing (Part 2).
- **Virtual Impedance:** Activated for fault current limiting (Part 5).

6.1.3 Islanding Detection Logic

IEEE 1547-2018 requires anti-islanding protection for all distributed energy resources. The PCS 3440 implements a hybrid detection scheme combining passive and active methods:

Passive Detection — Voltage and Frequency Thresholds

Monitor terminal voltage magnitude $|V_{PCC}|$ and frequency f_{PCC} against IEEE 1547-2018 Table 1 limits:

Table 6.1 Islanding Detection Voltage and Frequency Thresholds (IEEE 1547-2018)

Condition	Threshold	Trip Time	PSCAD Implementation
Over-voltage 2	$V > 1.20$ pu	0.16 s	Comparator, timer block
Over-voltage 1	$V > 1.10$ pu	2.0 s	Comparator, timer block
Under-voltage 1	$V < 0.88$ pu	2.0 s	Comparator, timer block
Under-voltage 2	$V < 0.50$ pu	0.16 s	Comparator, timer block
Over-frequency	$f > 60.5$ Hz (50 Hz: 51 Hz)	0.16 s	Comparator, timer block
Under-frequency	$f < 59.0$ Hz (50 Hz: 49 Hz)	0.16 s	Comparator, timer block

Active Detection — Rate of Change of Frequency (ROCOF)

ROCOF detects sudden frequency changes that indicate grid disconnection. Compute ROCOF as the numerical derivative of the filtered frequency signal:

$$ROCOF = \frac{df}{dt} = \frac{f_{PLL}(t) - f_{PLL}(t - T_{ROCOF})}{T_{ROCOF}}$$

Where $T_{ROCOF} = 20$ ms (one cycle averaging window). The ROCOF threshold for islanding detection:

$$|ROCOF| > ROCOF_{threshold} = 0.5 \text{ Hz/s} \Rightarrow \text{Potential island event}$$

PSCAD implementation: Use a derivative block with 20 ms averaging window, followed by an absolute value block and comparator with threshold 0.5 Hz/s.

Phase Jump Detection

A sudden phase angle jump of the grid voltage (compared to the PLL angle) indicates grid reconfiguration or islanding. The phase jump detector computes:

$$\Delta\theta = \theta_{PLL} - \theta_{VSM}$$

When operating grid-connected with both GFL and GFM modes active in parallel tracking, the angle difference is near-zero. A phase jump exceeding $\pm 15^\circ$ (± 0.262 rad) within a 10 ms window triggers an islanding flag:

$$|\Delta\theta_{jump}| = |\theta_{PLL,now} - \theta_{PLL,10ms_ago}| > 15^\circ \Rightarrow \text{Phase jump detected}$$

6.1.4 Combined Islanding Detection Logic (PSCAD)

PSCAD Islanding Detection Logic Gates

```
ISLAND_FLAG =
(OV2_Trip) OR (UV2_Trip)           [Extreme V – immediate]
OR (OV1_Trip AND Timer1_Elapsed)   [Moderate OV – 2s delay]
OR (UV1_Trip AND Timer2_Elapsed)   [Moderate UV – 2s delay]
OR (OF_Trip AND Timer3_Elapsed)    [Over-frequency – 0.16s]
OR (UF_Trip AND Timer4_Elapsed)    [Under-frequency – 0.16s]
OR (ROCOF_Flag AND ROCOF_Timer)    [ROCOF > 0.5 Hz/s for 0.1s]
OR (Phase_Jump_Flag)               [Phase jump > 15deg immediate]
```

Library components: CSMF > Logic > AND Gate, OR Gate; CSMF > Time Delay; CSMF > Comparator.

6.2 Bumpless Transfer and State Machine

6.2.1 State Machine for Mode Transition

The mode transition is governed by a finite state machine (FSM) with four states:

Table 6.2 Mode Transition State Machine

State	Name	Active Control	Transition Condition
S0	GRID-FOLLOWING	GFL (PLL + current control)	ISLAND_FLAG = 1 → S1

S1	TRANSITION-TO-GFM	Both GFL and GFM active; outputs averaged	VSM synced to load voltage → S2
S2	GRID-FORMING (ISLAND)	GFM (VSM + voltage control)	Grid restored AND GRID_OK = 1 → S3
S3	TRANSITION-TO-GFL	Both active; GFL PLL syncing to grid angle	PLL locked ($ \Delta\theta < 1^\circ$) → S0

6.2.2 Bumpless Transfer Mechanism

Bumpless transfer ensures that the control output continuity is maintained during the mode switch. The key challenge is that the GFL mode outputs a current reference while the GFM mode outputs a voltage reference. The bumpless mechanism:

Bumpless Transfer Algorithm

GFL → GFM Transition (State S0 → S1 → S2):

1. When ISLAND_FLAG triggers, the VSM integrator state ($\omega_{VSM}, \theta_{VSM}$) is initialized to the last PLL output: $\omega_{VSM,init} = \omega_{PLL}, \theta_{VSM,init} = \theta_{PLL}$.
2. The voltage controller integrator states are initialized to the pre-transition inverter modulation index.
3. Over a 50 ms ramp, the control mode switch S_{mode} transitions from 0 (GFL) to 1 (GFM): $S_{mode}(t) = \min(1, (t - t_{switch})/0.05)$.
4. The inverter output is a weighted sum: $u_{PWM} = (1 - S_{mode}) \cdot u_{GFL} + S_{mode} \cdot u_{GFM}$.

GFM → GFL Transition (State S2 → S3 → S0):

1. PLL is activated and locks to the restored grid voltage at PCC.
2. When $|\theta_{PLL} - \theta_{VSM}| < 1^\circ$ and $|\omega_{PLL} - \omega_{VSM}| < 0.001$ pu, transition is permitted.
3. Similar 50 ms weighted ramp from GFM to GFL control outputs.
4. Grid breaker is closed only after mode transition completes (to avoid large inrush).

6.2.3 PSCAD Implementation of State Machine and Bumpless Transfer

PSCAD State Machine Components

1. **State Register** (CSMF > Memory > Discrete Integrator with Reset): Holds current state S0–S3. Integer value 0–3.
2. **Transition Logic** (CSMF > Logic): AND/OR gates evaluate transition conditions. Outputs SET/RESET signals to state register.
3. **Ramp Generator** (CSMF > Integrator with limits 0/1 and rate = 20 pu/s = 0 → 1 in 50 ms): Generates $S_{mode}(t)$ ramp.
4. **Multiplexer Switches** (CSMF > Selector or Arithmetic Multiplier): Compute weighted sum $u = (1 - S_{mode}) \cdot u_{GFL} + S_{mode} \cdot u_{GFM}$ for each control channel (d-axis, q-axis).
5. **Integrator Initialization** (CSMF > Integrator with External IC input): VSM omega integrator accepts ω_{PLL} as initial condition when triggered by ISLAND_FLAG.
6. **Grid Breaker Control** (PSCAD > Circuit Breaker): Commanded by FSM output signal. Open in state S2; close in state S0 only after sync check passes.

Component:Ramp (Mode Switch S_mode)

Rate = 20.0 pu/s (0 to 1 in 50 ms) Limits: [0.0, 1.0]

Component:Comparator (Phase Sync Check)

Threshold = 0.01745 rad (= 1°) Input: $|\theta_{PLL} - \theta_{VSM}|$

Component:Timer (ROCOF delay) Time = 0.1 s

Component:Timer (UV/OV delay) Time = 2.0 s or 0.16 s (per IEEE 1547)

PART 7 — COMPLETE PSCAD SCHEMATIC

Part 7: Complete PSCAD Schematic

7.1 Power Stage and Filter Design

The complete PSCAD schematic for the PCS 3440 grid-forming inverter is organized into four hierarchical modules within PSCAD V5.0.2: (1) the AC power circuit, (2) the grid connection module, (3) the GFM control subsystem, and (4) the measurement and monitoring block. This hierarchical organization is recommended for clarity and reusability in PSCAD V5.0.2.

7.1.1 Three-Phase Inverter Power Stage (PSCAD)

The three-phase two-level voltage source inverter (2L-VSI) is modeled using PSCAD's Universal Bridge component or individual IGBT switches. The recommended approach for GFM studies is the averaged (switching function) model, which is computationally efficient and captures the fundamental control dynamics without resolving individual switching events:

Power Stage PSCAD Components

1. **DC Voltage Source** (Library: Master Library > Sources > Voltage Source, DC): $V_{DC} = 1200$ V. Internal resistance $R_{DC} = 0.001$ pu. Label: *VDC_Source*.
2. **DC Bus Capacitor** (Library: Master Library > Passive > Capacitor): $C_{DC} = 10$ mF (= 0.435 pu). Models DC bus energy storage.
3. **Three-Phase Voltage Source Converter (VSC)** (Library: Master Library > Machines > Universal Bridge, 3-phase, 6-pulse): Select "Firing pulses from gate signals" mode. Switching frequency: 5000 Hz carrier for detailed model. Or use "Averaged Model" (Library: FACTS > VSC Average Model) for control studies — recommended.
4. **Averaged VSC Model (Preferred)**: The PSCAD "Voltage-Sourced Converter (Averaged)" block accepts per-unit modulation index m_a and angle θ inputs, outputting three-phase voltages at the AC terminals. Set base: 3440 kVA, 690 V.

7.1.2 LC Output Filter (PSCAD Components)

The LC filter is connected between the VSC AC terminals and the Point of Common Coupling (PCC) bus:

Table 7.1 LC Filter PSCAD Component Specifications

Component	PSCAD Library Path	Value	Per-Unit	PSCAD Label
Filter Inductor	Master Library > Passive > Inductor (3-phase coupled)	0.088 mH/phase	0.200 pu	Lf_filter
Inductor Resistance	Embedded in inductor (R field)	1.384 mΩ	0.010 pu	Rf_filter
Filter Capacitor	Master Library > Passive > Capacitor (3-phase, Y-grounded)	1.150 mF/phase	0.050 pu	Cf_filter
Cap. Damping R	Series with Cf (Master Library > Passive > Resistor)	0.277 mΩ	0.002 pu	Rd_damping

Grid Inductance	Master Library > Passive > Inductor	0.044 mH	0.100 pu	Lg_grid
Grid Resistance	Embedded in grid inductor	0.692 mΩ	0.005 pu	Rg_grid

7.1.3 Grid Connection Interface

The grid connection interface includes measurement nodes, protection breakers, and an ideal voltage source representing the utility grid:

- **PCC Bus Node:** The junction between the LC filter output, local load, and grid connection breaker. All three measurement blocks (V, I, P, Q) are connected here.
- **Grid Circuit Breaker** (Library: PSCAD > CSMF > Breaker > 3-Phase Breaker): Controls grid connection. Commanded by FSM output S_{grid} . Timing accuracy: ± 0.5 ms.
- **Load Resistor/Inductor** (optional for island mode testing): Passive R-L load at PCC to simulate local island load. $R_{load} = 1.0$ pu, $L_{load} = 0.3$ pu.
- **Grid Voltage Source** (Library: PSCAD > Sources > 3-Phase Voltage Source): $V_{grid} = 690$ V (L-L), $f = 50$ Hz. Internal impedance: $Z_{grid} = 0.05$ pu (stiff grid equivalent).

7.2 Detailed PSCAD Component List

The following table provides a complete, reference-grade list of all PSCAD components required to build the PCS 3440 grid-forming inverter model in PSCAD V5.0.2, with library paths, parameter values, and signal connections.

Table 7.2 Complete PSCAD Component List — PCS 3440 GFM System

Block Name	PSCAD Library	Key Parameters	Input Signals	Output Signals
DC Source	Sources > DC Voltage	V=1200V, R=0.138mΩ	—	VDC bus
VSC Averaged	FACTS > VSC Average	3.44MVA, 690V base	m_a, theta, VDC	V_abc (AC)
Lf (3-phase)	Passive > Inductance	L=0.088mH, R=1.384mΩ	VSC output	Filter current I_L
Cf (Y-grounded)	Passive > Capacitor	C=1150μF (per phase)	LC midpoint	Cap voltage V_C
V Meas (3ph)	Meters > Voltage	Output in pu (base 690V)	PCC bus	Vabc_pu

I Meas (3ph)	Meters > Current	Output in pu (base 2879A)	Filter inductor	Iabc_pu
P,Q Meter	Meters > Power	Base 3.44MVA	V, I signals	P_pu, Q_pu
Park Transform (abc → dq)	CSMF > Transforms	3-phase to dq	Vabc or Iabc, theta	Vd, Vq or Id, Iq
Park Transform (dq → abc)	CSMF > Transforms	dq to 3-phase	Vd*, Vq*, theta	Vabc* (modulator)
PI (Voltage d-ax)	CSMF > Controllers > PI	Kp=23.55, Ki=1109	Vd*-Vd_meas	Id_ref
PI (Voltage q-ax)	CSMF > Controllers > PI	Kp=23.55, Ki=1109	Vq*-Vq_meas	Iq_ref
PI (Current d-ax)	CSMF > Controllers > PI	Kp=942.5, Ki=47.12	Id*-Id_meas	Vd_PWM
PI (Current q-ax)	CSMF > Controllers > PI	Kp=942.5, Ki=47.12	Iq*-Iq_meas	Vq_PWM
Integrator (omega)	CSMF > Integrator	IC=1.0, Lim [0.95, 1.05]	d_omega/dt	omega_pu
Integrator (theta)	CSMF > Integrator	IC=0, No limits	omega_phys	theta_VSM
LP Filter (P meas)	CSMF > Filters > 1st Order	T=15.92ms	P_pu	P_filt
LP Filter (Q meas)	CSMF > Filters > 1st Order	T=31.83ms	Q_pu	Q_filt
Gain (m_P droop)	CSMF > Gain	K=0.05	DeltaP	Delta_omega
Gain (m_Q droop)	CSMF > Gain	K=0.10	DeltaQ	Delta_V
Gain (1/2H)	CSMF > Gain	K=0.125	P_net	d_omega/dt
Gain (D_vsm)	CSMF > Gain	K=20.0	omega_error	P_damping
Grid Breaker (3ph)	PSCAD > Breaker	3-phase, timed open/close	FSM S_grid signal	Grid connected/disconnected

DSOGI-PLL	Machine/Control PLL	>	BW=30Hz, nominal=50Hz	V_PCC_abc	theta_PLL, omega_PLL
Limiters (all)	CSMF > Limiters		Per-block specifications	Unbounded signals	Clamped signals
Oscilloscope Output	/ PSCAD Signals	> Output >	Record: V, I, P, Q, f, theta	Any signal	Time-domain plot data

7.2.1 Simulation Parameters (PSCAD V5.0.2 Project Setup)

Simulation Time Step: $\Delta t = 50\ \mu s$ (20× switching period / 400 steps per 50Hz cycle)
Total Simulation Time:5.0 s (covers startup, steady-state, and island transition)
Output Sample Rate:10 kHz (every 10 time steps)
Solver:EMTDC trapezoidal (default)
Frequency Base:50 Hz
Power Base:3440 kVA
Voltage Base (AC):690 V (L-L)
Startup Sequence: Pre-charge DC bus at t=0, close grid breaker at t=0.1s, ramp power to set-point over 0.5s

PART 8 — DETAILED CALCULATIONS

Part 8: Detailed Calculations (Step-by-Step Per-Unit)

This section consolidates all per-unit calculations from Parts 1–7 into a single reference, showing each derivation step by step. All values are normalized to the base system: $S_{base} = 3.44\text{ MVA}$, $V_{base} = 690\text{ V (L-L)}$, $f_{base} = 50\text{ Hz}$.

8.1 Complete Base Value Derivation Chain

Table 8.1 Step-by-Step Base Value Calculations

Step	Quantity	Formula	Calculation	Result
1	S_{base}	Nameplate	3.44×10^6	3,440,000 VA
2	$V_{base,LL}$	Nameplate	690 V	690.0 V
3	$V_{base,LN}$	$V_{LL}/\sqrt{3}$	$690/1.7321$	398.37 V
4	I_{base}	$S/(\sqrt{3}V_{LL})$	$3,440,000/(1.7321 \times 690)$	2,879.0 A

5	Z_{base}	V_{LL}^2/S	$690^2/3,440,000$	0.13840 Ω
6	ω_{base}	$2\pi f$	$2\pi \times 50$	314.159 rad/s
7	L_{base}	Z/ω	$0.13840/314.159$	0.4404 mH
8	C_{base}	$1/(\omega Z)$	$1/(314.159 \times 0.13840)$	23.00 mF
9	T_{base}	$1/\omega_{base}$	$1/314.159$	3.183 ms

8.2 Droop Coefficient Calculations

Step 1: Define droop percentage. 5% droop means a 5% change in frequency per 100% change in power.

$$D\% = 5\% \Rightarrow \frac{\Delta\omega_{pu}}{\Delta P_{pu}} = \frac{0.05 \times 1.0}{1.0} = 0.05 \text{ pu/pu}$$

Step 2: P-f droop coefficient.

$$m_P = 0.05 \text{ pu} \Rightarrow K_{droop,P} = 0.05 \text{ (PSCAD Gain block value)}$$

Step 3: Physical droop slope in Hz/MW.

$$k_P^{phys} = \frac{0.05 \times 50 \text{ Hz}}{3.44 \text{ MW}} = \frac{2.5}{3.44} = 0.7267 \text{ Hz/MW}$$

Step 4: Q-V droop coefficient. 5% voltage change at $Q_{max} = 0.5$ pu:

$$m_Q = \frac{0.05}{0.5} = 0.10 \text{ pu/pu} \Rightarrow K_{droop,Q} = 0.10$$

Step 5: P filter time constant. 10 Hz low-pass filter:

$$T_{fP} = \frac{1}{2\pi \times 10} = \frac{1}{62.83} = 15.92 \text{ ms}$$

Step 6: Q filter time constant. 5 Hz low-pass filter:

$$T_{fQ} = \frac{1}{2\pi \times 5} = \frac{1}{31.42} = 31.83 \text{ ms}$$

8.3 VSM Parameter Calculations

Step 1: Per-unit inertia gain for swing equation integrator.

$$\frac{1}{2H} = \frac{1}{2 \times 4} = \frac{1}{8} = 0.125 \text{ pu}$$

Step 2: Virtual inertia in kg·m².

$$J = \frac{2H \cdot S_{base}}{\omega_0^2} = \frac{8 \times 3,440,000}{314.159^2} = \frac{27,520,000}{98,715.9} = 278.76 \text{ kg} \cdot \text{m}^2$$

Step 3: Damping power at 0.01 pu frequency deviation.

$$\Delta P_{damp} = D \cdot \Delta\omega = 20 \times 0.01 = 0.20 \text{ pu} = 688 \text{ kW}$$

Step 4: VSM response time constant.

$$\tau_{VSM} = \frac{2H}{D} = \frac{8}{20} = 0.40 \text{ s} = 400 \text{ ms}$$

Step 5: VSM governor bandwidth.

$$f_{BW,VSM} = \frac{1}{2\pi\tau_{VSM}} = \frac{1}{2\pi \times 0.40} = 0.398 \text{ Hz}$$

8.4 Virtual Impedance Calculations

Step 1: Maximum fault current limit.

$$I_{fault,limit} = 1.5 \text{ pu} = 1.5 \times 2,879 = 4,319 \text{ A}$$

Step 2: Required virtual impedance magnitude.

$$|Z_{virt}| = \frac{V_0}{I_{limit}} - Z_{grid} = \frac{1.0}{1.5} - 0.05 = 0.617 \text{ pu}$$

Step 3: R and X decomposition with X/R = 10.

$$R_{virt} = \frac{|Z_{virt}|}{\sqrt{1 + 100}} = \frac{0.617}{10.05} = 0.0614 \text{ pu}$$

$$X_{virt} = 10 \times 0.0614 = 0.614 \text{ pu}$$

$$L_{virt} = \frac{X_{virt}}{\omega_0} = 0.614 \text{ pu} = 0.614 \times 0.4404 = 0.2704 \text{ mH}$$

8.5 PI Controller Gain Calculations

Voltage Controller (Outer Loop):

$$K_{pV} = \omega_{BW,V} \times C_{f,pu} = 471 \times 0.050 = 23.55 \text{ pu}$$

$$T_{iV} = \frac{10}{\omega_{BW,V}} = \frac{10}{471} = 21.23 \text{ ms}$$

$$K_{iV} = \frac{K_{pV}}{T_{iV}} = \frac{23.55}{0.02123} = 1,110 \text{ pu/s}$$

Current Controller (Inner Loop):

$$K_{pI} = \omega_{BW,I} \times L_{f,pu} = 4,712 \times 0.200 = 942.5 \text{ pu}$$

$$T_{iI} = \frac{L_{f,pu}}{R_{f,pu}} = \frac{0.200}{0.010} = 20.0 \text{ pu} \cdot \text{s}$$

$$K_{iI} = \frac{K_{pI}}{T_{iI}} = \frac{942.5}{20.0} = 47.12 \text{ pu/s}$$

8.6 Filter Parameter Calculations

LC filter resonance frequency:

$$f_{res} = \frac{1}{2\pi\sqrt{L_f C_f}} = \frac{1}{2\pi\sqrt{0.088 \times 10^{-3} \times 1.150 \times 10^{-3}}} = \frac{1}{2\pi \times 0.318 \times 10^{-3}} = 500 \text{ Hz}$$

Switching frequency to resonance ratio:

$$\frac{f_{sw}}{f_{res}} = \frac{5000}{500} = 10 \Rightarrow \text{Harmonic attenuation at } f_{sw} : \left(\frac{f_{res}}{f_{sw}} \right)^2 = (0.1)^2 = 0.01$$

Capacitor damping resistance (passive damping):

$$R_d = \frac{1}{3} \sqrt{\frac{L_f}{C_f}} = \frac{1}{3} \sqrt{\frac{0.088 \times 10^{-3}}{1.150 \times 10^{-3}}} = \frac{1}{3} \sqrt{0.0765} = \frac{0.2766}{3} = 0.0922 \text{ } \Omega$$

$$R_{d,pu} = \frac{R_d}{Z_{base}} = \frac{0.0922}{0.1384} = 0.002 \text{ pu}$$

Table 8.2 Complete Parameter Summary for PCS 3440 GFM Control

Parameter	Symbol	Value (pu)	Physical
P-f Droop Coeff.	m_P	0.05	0.7267 Hz/MW
Q-V Droop Coeff.	m_Q	0.10	0.0291 kV/kVAr
VSM Inertia 2H	M	8.00	278.76 kg·m ²

VSM Damping	D	20.00	688 kW/0.5 Hz
Voltage PI Kp	K_{pV}	23.55	—
Voltage PI Ki	K_{iV}	1,110	—
Current PI Kp	K_{pI}	942.5	—
Current PI Ki	K_{iI}	47.12	—
Virtual R	R_{virt}	0.0614	8.50 mΩ
Virtual L	L_{virt}	0.614	0.270 mH
Filter L	L_f	0.200	0.088 mH
Filter C	C_f	0.050	1.150 mF
Filter Res. Freq.	f_{res}	—	500 Hz

PART 9 — STEP-BY-STEP IMPLEMENTATION GUIDE

Part 9: Step-by-Step PSCAD V5.0.2 Implementation Guide

This section provides a complete, sequential procedure for building the PCS 3440 GFM simulation in PSCAD V5.0.2, suitable for an engineer with basic PSCAD familiarity. Steps are organized in logical build order, from project creation through to simulation verification.

9.1 Creating a New PSCAD V5.0.2 Project

Step 1: Project Setup

1. Launch PSCAD V5.0.2. Select **File > New > Project**. Name: *PCS3440_GFM_V1*. Save to dedicated folder.
2. Open **Project Settings** (right-click project in workspace tree). Set:
 - Duration: 5.0 s
 - Time Step: 50.0 μ s (0.000050 s)
 - Plot Step: 200.0 μ s (sample rate: 5 kHz for smooth waveforms)
 - Channels: Select all measurement signals for recording
 - Frequency: 50 Hz
3. In **Settings > Solver**: Confirm EMTDC trapezoidal integration. Tolerances: default.
4. Create four Page canvases: "Power_Circuit", "GFM_Control", "Measurement", "Protection_Transitions". Use **Insert > New Page**.

9.2 Building the Power Stage

Step 2: Place Power Components on Power_Circuit Page

1. **DC Source:** From Master Library > Sources, drag a DC Voltage Source onto the canvas. Double-click to set: $V = 1200$ V. Add series resistance 0.138 m Ω .
2. **DC Bus Capacitor:** Master Library > Passive > Capacitor. $C = 10000$ μ F. Connect between positive DC bus and ground node.
3. **VSC Block:** From FACTS Library, drag "VSC (Averaged)" block. Parameters: Rated VA = $3440e3$, $V_{base} = 690$ V. Connect DC+/DC- terminals to DC bus. AC terminals to filter input.
4. **Filter Inductor (3-phase):** Master Library > Passive > Single-Phase RL (place three in parallel). Each phase: $L = 88$ μ H, $R = 1.384$ m Ω . Connect VSC AC output $\rightarrow$ Lf $\rightarrow$ midpoint node (LP1).
5. **Filter Capacitor (3-phase, Y):** Place three capacitors from midpoint LP1 to neutral. Each: $C = 1150$ μ F. Add series damping $R = 0.0922$ Ω in series with each capacitor.
6. **Grid Inductor:** Single-phase RL, three phases: $L = 44$ μ H, $R = 0.692$ m Ω . Connect LP1 $\rightarrow$ Lg $\rightarrow$ PCC bus.
7. **Grid Breaker (3-phase):** PSCAD Master Library > Breaker. Set "Timed Breaker" mode. Control signal: *BRK_Grid* from FSM on Protection page. Initial state: Closed.
8. **Grid Voltage Source:** Sources > 3-Phase Voltage Source (AC). $V_{LL} = 690$ V, $f = 50$ Hz. Internal $R = 0.0069$ Ω (0.05 pu). Connect after grid breaker.
9. **Load (for island testing):** Three single-phase R-L branches from PCC to neutral: $R = 0.1384$ Ω (1.0 pu), $L = 132$ μ H (0.3 pu). Place a switch (Load_SW) to connect/disconnect during tests.

9.3 Building the GFM Control Subsystem

Step 3: GFM Control Block Construction (GFM_Control Page)

3A — P-f and Q-V Droop Blocks:

1. Place LP Filter block ($T = 15.92$ ms) — input: P_{pu} from meter. Output: P_{filt} .
2. Summing Junction: inputs $+P_{ref}$ (set to 1.0 pu), $-P_{filt}$. Output: ΔP .
3. Gain block ($K = 0.05$): input ΔP . Output: $\Delta \omega$.
4. Summing Junction: inputs $+\omega_{nom}$ (1.0 pu), $-\Delta \omega$. Output: ω_{droop} .
5. Hard Limiter (0.95 to 1.05): input ω_{droop} . Output: P_m (virtual mechanical power freq.).
6. Repeat steps 1–5 for Q-V droop: $T=31.83$ ms, $K=0.10$, output: V_{star} (clamped 0.88–1.10 pu).

3B — VSM Swing Equation Blocks:

1. LP Filter ($T=10\text{ms}$): input P_e (measured). Output: P_{e_filt} .
2. Summing Junction ($+P_m, -P_{e_filt}$): Output ΔP_{net} .
3. Gain ($K=20.0$, label D_{vsm}): input ω_{error} ($= \omega_{pu} - 1.0$). Output: P_{damp} .
4. Summing Junction ($+\Delta P_{net}, -P_{damp}$): Output P_{acc} .
5. Gain ($K=0.125$, label 1_{over_2H}): input P_{acc} . Output: d_{ω_dt} .
6. Integrator ($IC=1.0$, $lim=[0.95, 1.05]$): input d_{ω_dt} . Output: ω_{pu} . **Feed ω_{pu} back to step 3 as $\omega_{error} = \omega_{pu} - 1.0$.**
7. Gain ($K=314.159$): input ω_{pu} . Output: ω_{rad} .
8. Integrator ($IC=0$, no limits): input ω_{rad} . Output: θ_{raw} .
9. Modulo ($\text{mod } 2\pi = 6.2832$): input θ_{raw} . Output: θ_{VSM} .

3C — Voltage and Current PI Controllers:

1. Park Transform ($abc \rightarrow dq$, $\text{angle}=\theta_{VSM}$): input V_{c_abc} . Outputs: V_{c_d} , V_{c_q} .
2. PI ($K_p=23.55$, $K_i=1109$, $lim=\pm 1.10$): $\text{error} = V_{star} - V_{c_d}$. Output: I_{d_ref} .
3. PI ($K_p=23.55$, $K_i=1109$, $lim=\pm 1.10$): $\text{error} = 0 - V_{c_q}$. Output: I_{q_ref} .
4. Park Transform ($abc \rightarrow dq$, $\text{angle}=\theta_{VSM}$): input I_{L_abc} . Outputs: I_{L_d} , I_{L_q} .
5. PI ($K_p=942.5$, $K_i=47.12$, $lim=\pm 1.15$): $\text{error} = I_{d_ref} - I_{L_d}$. Output: V_{d_ref} .
6. PI ($K_p=942.5$, $K_i=47.12$, $lim=\pm 1.15$): $\text{error} = I_{q_ref} - I_{L_q}$. Output: V_{q_ref} .
7. Add cross-coupling: $V_{d_ref} += \omega_{pu} \times L_{f_pu} \times I_{L_q}$; $V_{q_ref} -= \omega_{pu} \times L_{f_pu} \times I_{L_d}$.
8. Park Transform ($dq \rightarrow abc$, $\text{angle}=\theta_{VSM}$): input V_{d_ref} , V_{q_ref} . Outputs: modulation signals V_{abc_ref} .
9. Divide V_{abc_ref} by $(V_{DC}/2)$ to normalize to modulation index m_a (range ± 1.0).
10. Feed m_a and θ_{VSM} to VSC Averaged block inputs on Power_Circuit page.

9.4 Building the Virtual Impedance Block

Step 4: Virtual Impedance Implementation

1. Math: compute $|I| = \sqrt{IL_d^2 + IL_q^2}$ using Multiply and Sqrt blocks.
2. Arithmetic: $k_{virt} = \text{clip}((|I| - 1.0)/0.5, 0, 1)$. Use Comparators, Subtraction, Division, and a Hard Limiter [0,1].
3. $R_{eff} = k_{virt} \times 0.0614$ (Gain block $K=0.0614 \times k_{virt}$ via Multiplier).
4. $L_{eff} = k_{virt} \times 0.614$ (Gain block $K=0.614 \times k_{virt}$ via Multiplier).
5. $dVd = R_{eff} \times IL_d - \omega_{pu} \times L_{eff} \times IL_q$ (two Multipliers + Subtraction).
6. $dVq = R_{eff} \times IL_q + \omega_{pu} \times L_{eff} \times IL_d$ (two Multipliers + Addition).
7. Subtract: $Vd_{ref_final} = Vd_{ref} - dVd$; $Vq_{ref_final} = Vq_{ref} - dVq$.
8. Use Vd_{ref_final} and Vq_{ref_final} as inputs to the $dq \rightarrow abc$ Park Transform.

9.5 Connecting Measurement and Feedback Signals

PSCAD uses named signals ("wires") for cross-page connections. Use **Insert > Signal Label** to create named output points, and **Insert > Signal Reference** to receive them on other pages:

- **V_PCC_abc**: 3-phase voltage measurement at PCC → send to GFM_Control (Park transform, PLL), Protection_Transitions (islanding detection).
- **IL_abc**: Inductor current → GFM_Control (Park transform, virtual impedance magnitude).
- **P_pu, Q_pu**: Power meter outputs → GFM_Control (droop filters), Measurement page (oscilloscope).
- **theta_VSM, omega_pu**: VSM outputs → all Park transforms, Protection_Transitions (ROCOF, phase-jump detection).
- **theta_PLL, omega_PLL**: PLL outputs → Protection_Transitions (bumpless transfer IC), GFL control path.
- **BRK_Grid**: Grid breaker control signal from FSM → Power_Circuit breaker control.

9.6 Setting Component Parameters — Quick Reference Checklist

Table 9.1 Parameter Setting Checklist for PSCAD V5.0.2

Component	Parameter	Value to Set	Verified?
VSC Averaged	Rated MVA / V_base	3.44 MVA / 690 V	☐

Lf per phase	L / R	88 μ H / 1.384 m Ω	☐
Cf per phase	C / Rd	1150 μ F / 92.2 m Ω	☐
LP Filter (P)	Time constant T	15.92 ms	☐
LP Filter (Q)	Time constant T	31.83 ms	☐
Gain (m_P)	Gain K	0.05	☐
Gain (m_Q)	Gain K	0.10	☐
Gain (1/2H)	Gain K	0.125	☐
Gain (D_vsm)	Gain K	20.0	☐
Integrator (omega)	IC / Limits	1.0 / [0.95, 1.05]	☐
Integrator (theta)	IC / Limits	0.0 / none	☐
PI Voltage (d,q)	Kp / Ki / Lim	23.55 / 1109 / $\pm$ 1.10	☐
PI Current (d,q)	Kp / Ki / Lim	942.5 / 47.12 / $\pm$ 1.15	☐
Limiter (omega_ref)	Min / Max	0.95 / 1.05	☐
Limiter (V_ref)	Min / Max	0.88 / 1.10	☐
Grid Voltage Source	V_LL / f	690 V / 50 Hz	☐

9.7 Testing and Verification Procedures

Verification is conducted in four sequential test scenarios, each building on the previous:

1. **Test 1 — GFM Island Mode Steady-State:** Run simulation with grid breaker open (island mode). Apply R-L load (R=1.0, L=0.3 pu). Verify at t=1.0s: V_PCC = 690 $\pm$ 5 V, f = 50.0 $\pm$ 0.25 Hz, P and Q stable. Check: voltage THD < 2%, no oscillations.
2. **Test 2 — Load Step Response:** At t=2.0s, step load from 0.5 to 1.0 pu. Observe: frequency nadir within VSM response (target: <-0.5 Hz dip), recovery within 1.5s, V_PCC deviation < $\pm$ 5%.
3. **Test 3 — Grid Reconnection (GFM $\rightarrow$ GFL):** At t=2.5s, restore grid (close breaker). Observe: PLL lock within 0.5s, smooth mode transition, P and Q ramp to grid-commanded setpoints.
4. **Test 4 — Islanding Detection:** At t=3.5s, open grid breaker externally (simulating island event). Verify ISLAND_FLAG triggers within 0.16s for voltage/frequency violations, or within 0.3s for ROCOF detection.

Common PSCAD Troubleshooting Tips

- **Oscillations at startup:** Check integrator initial conditions. Omega integrator IC must be exactly 1.0 pu; theta IC must match grid angle.
- **Modulation index saturation:** Verify V_{d_ref} and V_{q_ref} are within ± 1.15 pu. If clipping, check current PI limiter settings.
- **Algebraic loop warnings:** Enable "Output Delay = 1 time-step" on omega integrator. Rebuild solution sequence if PSCAD reports ordering errors.
- **Slow simulation:** Use Averaged VSC model (not switching model) for control studies. Reduce plot step to 500 μ s if real-time output is not needed.
- **Incorrect power base:** Ensure ALL meters, VSC, and PI controllers share the same pu base (3.44 MVA, 690 V). Mixed bases cause gain errors that are difficult to trace.

PART 10 — BLOCK DIAGRAMS AND SYSTEM ARCHITECTURE

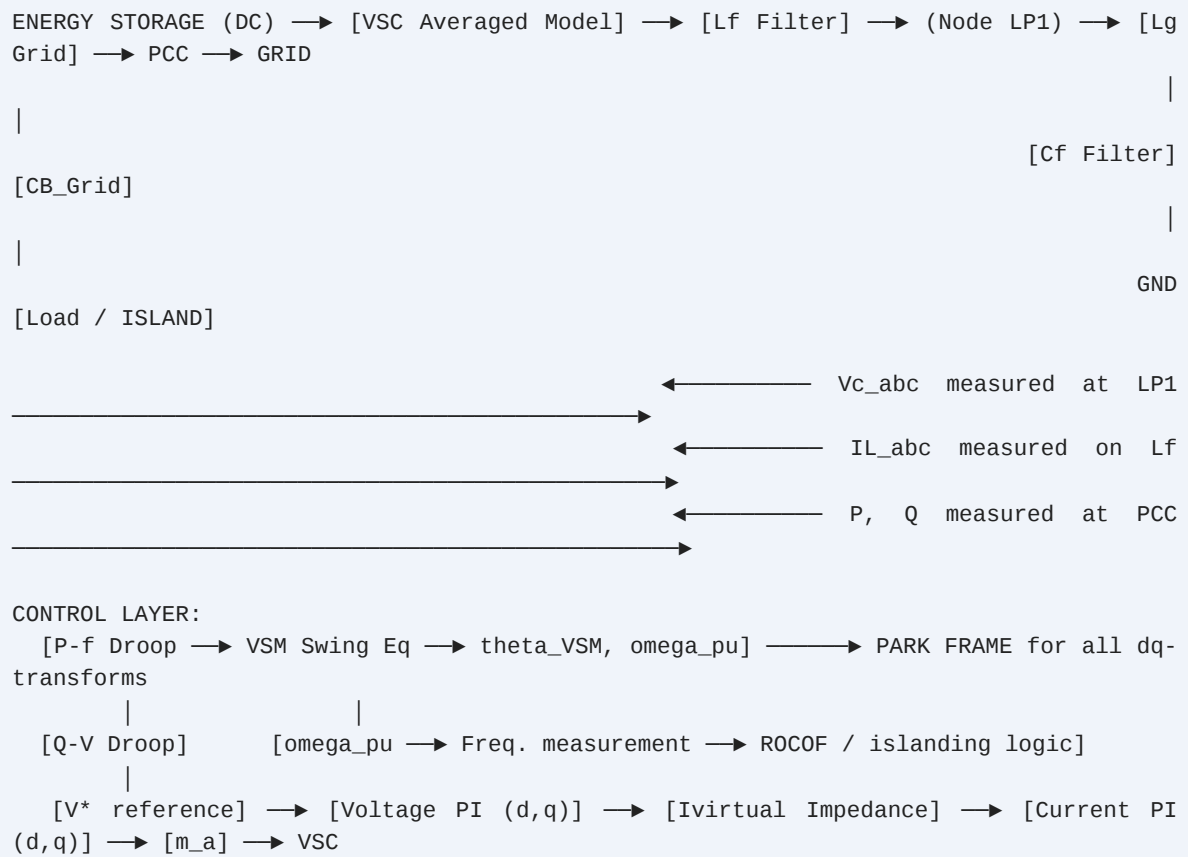
Part 10: Block Diagrams and System Architecture

This section presents the complete set of system architecture diagrams and block diagrams for the PCS 3440 GFM control system. All diagrams are rendered as structured SVG-equivalent HTML tables to provide precise, print-quality block diagrams compatible with the PDF output.

10.1 Overall System Architecture

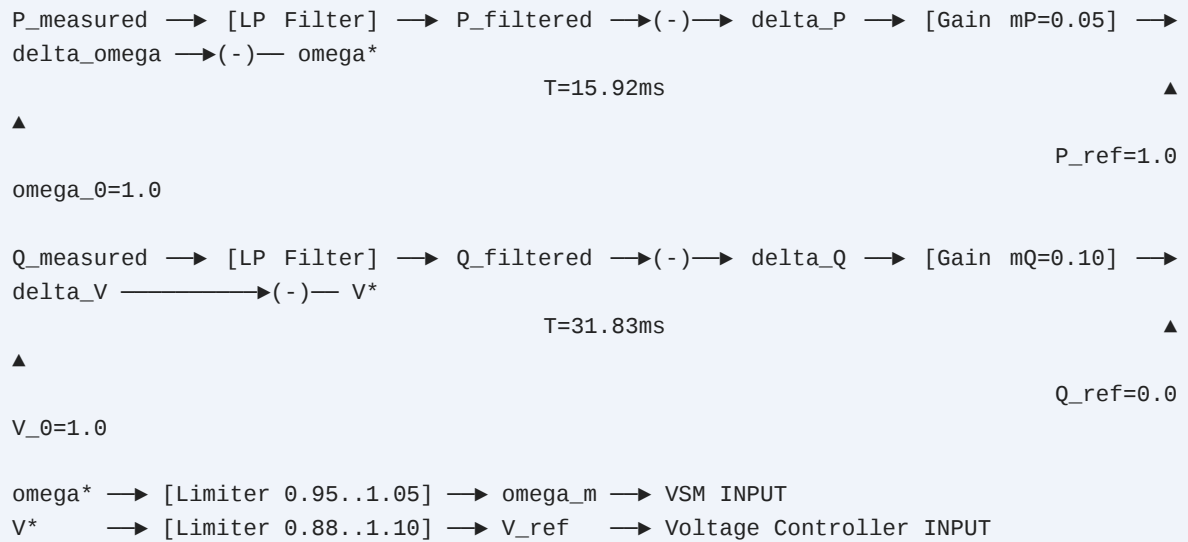
The PCS 3440 GFM system architecture encompasses four hierarchical layers: the physical AC/DC power stage, the innermost current control loop, the intermediate voltage control loop, and the outermost power/frequency/voltage regulation layer (droop + VSM). The complete signal flow is:

Signal Flow Architecture — Top-Level View



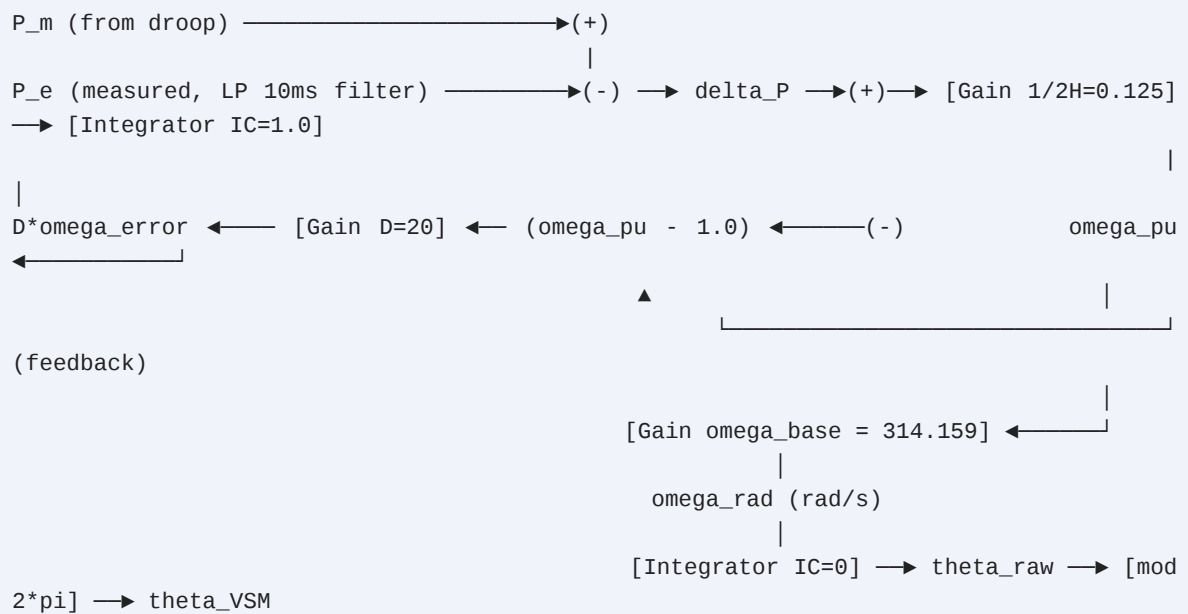
10.2 Droop Control Block Diagram

P-f Droop Block Diagram (Signal Flow)



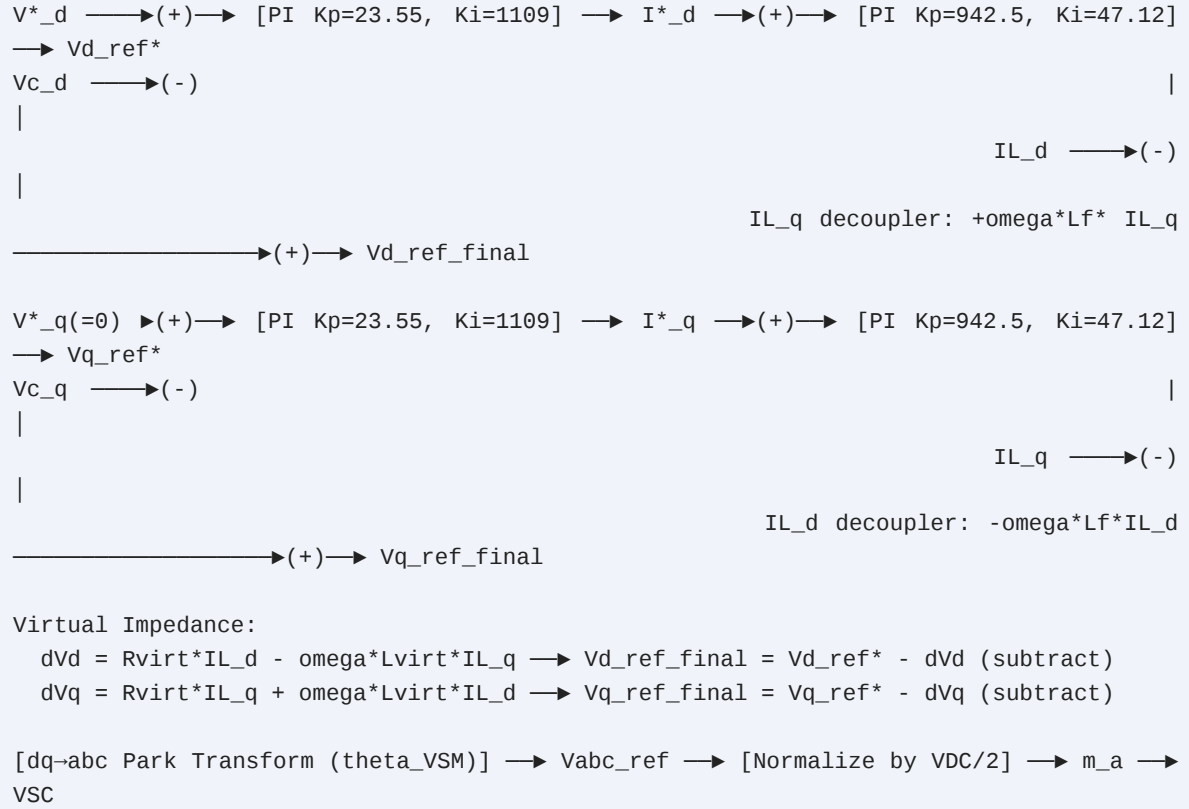
10.3 VSM Block Diagram

VSM Swing Equation Block Diagram



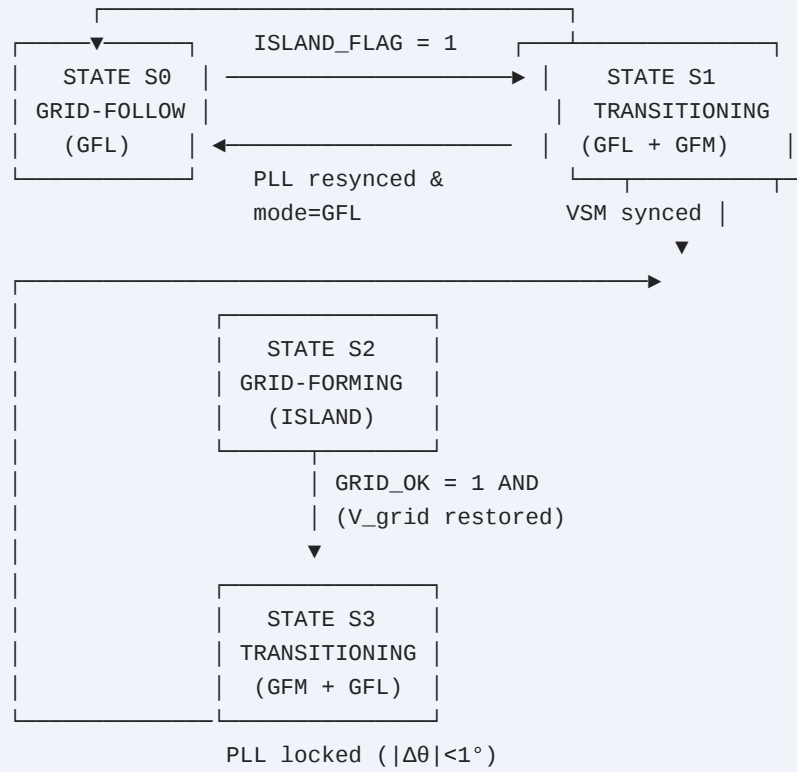
10.4 Voltage and Current Control Loop Diagram

Cascaded V-I Control Block Diagram (dq Frame)



10.5 Mode Transition State Diagram

Finite State Machine — Mode Transition Diagram



10.6 Summary of All Key Control Parameters

Table 10.1 Master Parameter Reference — PCS 3440 GFM at 690V / 3.44 MVA

Section	Parameter	Per-Unit Value	Physical Value	PSCAD Block
Base	Voltage (L-L)	1.000	690 V	VSC base setting
Base	Apparent Power	1.000	3.44 MVA	All meter bases
Base	Current	1.000	2,879 A	I-meter scale
Base	Impedance	1.000	0.1384 Ω	Z scale
Droop	m_P (P-f)	0.050	0.7267 Hz/MW	Gain block
Droop	m_Q (Q-V)	0.100	0.0291 kV/kVAr	Gain block

VSM	H (inertia)	4.00 s	278.76 kg·m ²	1/2H = 0.125
VSM	D (damping)	20.00	688 kW/0.5 Hz	Gain block
Voltage PI	K _p / K _i	23.55 / 1110	—	PI block
Current PI	K _p / K _i	942.5 / 47.12	—	PI block
Virtual Z	R / L	0.0614 / 0.614	8.5 mΩ / 0.27 mH	Gain × k _{virt}
Filter	L _f / C _f	0.200 / 0.050	88 μH / 1150 μF	Passive elements
Islanding	ROCOF threshold	—	0.5 Hz/s	Comparator
Transition	Ramp time	—	50 ms	Ramp generator
Simulation	Time step Δt	—	50 μs	Project settings

Standards Compliance Summary

The PCS 3440 GFM implementation described in this document is designed to comply with the following standards: **IEEE Std 1547-2018** (DER interconnection requirements, including voltage/frequency ride-through and islanding detection), **IEEE Std 2800-2022** (interconnection requirements for inverter-based resources), **IEC 62786:2017** (distributed energy resources interconnection with distribution networks), and **IEC TR 63227:2021** (virtual synchronous machine requirements for inverter-based resources). The virtual inertia ($H = 4$ s) and damping ($D = 20$) parameters are specifically chosen to meet grid code requirements for inertia response contribution under emerging European and North American grid codes that mandate minimum inertia constants for IBRs connected to transmission systems.